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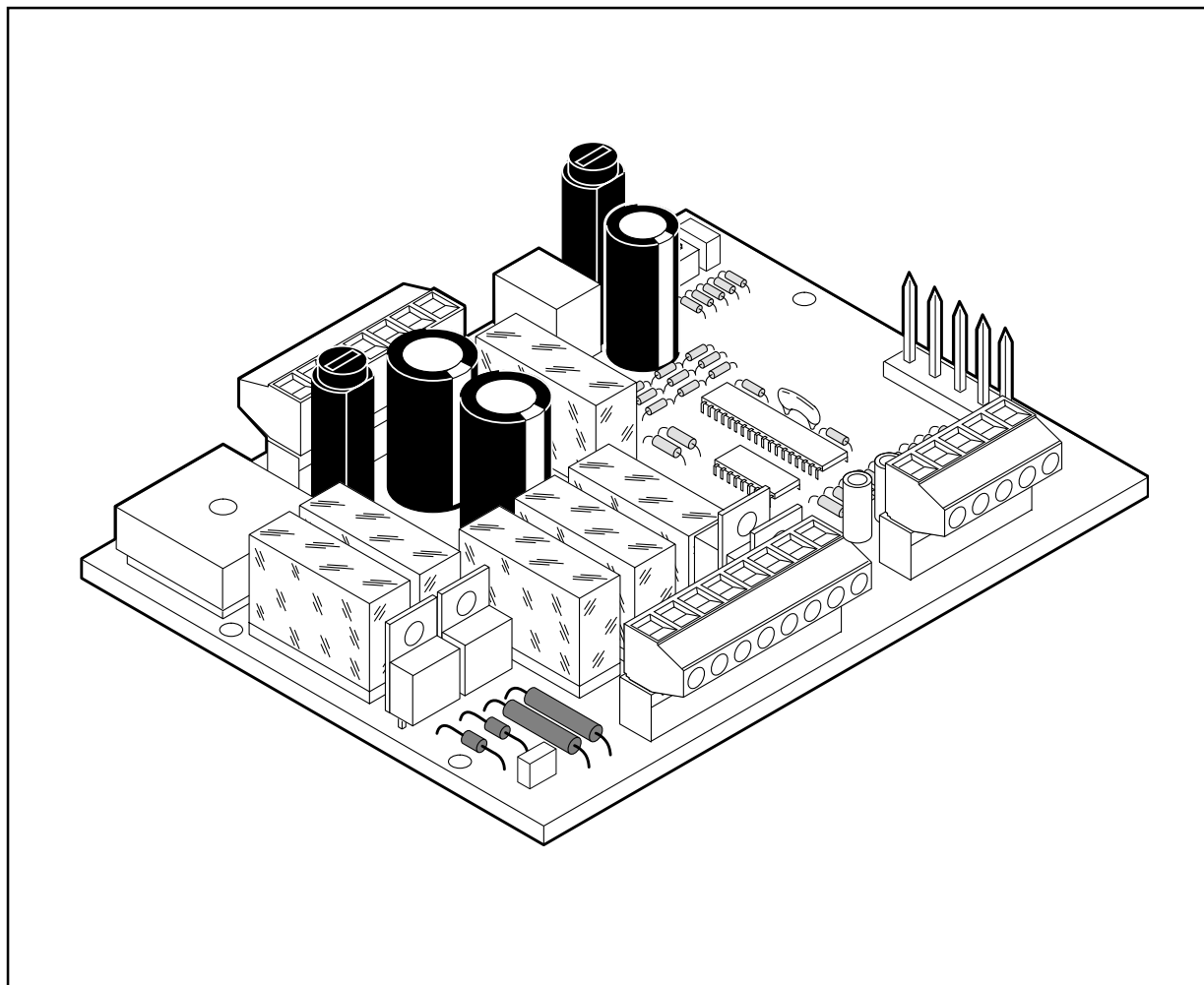
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424 MPS



FAAC



CE DECLARATION OF CONFORMITY

Manufacturer: FAAC S.p.A.

Address: Via Benini, 1 - 40069 Zola Predosa BOLOGNA - ITALY

Declares that: Electronic appliance 424MPS

- conforms to the essential safety requirements of the following directives:

73/23/EEC and subsequent amendment 93/68/EEC.

89/336/EEC and subsequent amendment 92/31/CEE and 93/68/EEC

Additional information:

This product underwent tests in a typical, uniform configuration (all products manufactured by FAAC S.p.A.).

Bologna 01 January 2000

The Managing Director
A. Bassi



WARNING FOR THE INSTALLER

GENERAL SAFETY OBLIGATIONS

- CAUTION! It is important for personal safety to follow all the instructions carefully. Incorrect installation or misuse of the product may cause people serious harm.**
- Read the instructions carefully before starting to install the product.
- Packaging material (plastic, polystyrene, etc.) must not be left within reach of children as it is a potential source of danger.
- Keep the instructions in a safe place for future reference.
- This product was designed and manufactured strictly for the use indicated in this documentation. Any other not expressly indicated use may damage the product and/or be a source of danger.
- FAAC accepts no responsibility due to improper use of the automation or use other than that intended.
- Do not install the appliance in an area subject to explosion hazard: inflammable gasses or fumes are a serious safety hazard.
- Mechanical construction elements must meet the provisions of UNI8612, CEN pr EN 12604 and CEN pr EN 12605 Standards.
To obtain an adequate level of safety in non EU countries, the above mentioned Standards must be observed in addition to national standards.
- FAAC will not accept responsibility if the principles of Good Workmanship are disregarded in constructing the closing elements to be motorised, and if any deformation occurs during use of the said elements.
- Installation must meet the following Standards: UNI8612, CEN pr EN 12453 and CEN pr EN 12635.
The safety class for the automation must be C+D.
- Before carrying out any work on the system, switch off the power supply.
- The mains power supply of the automation must be fitted with a all-pole switch with contact opening distance of 3mm or greater. Use of a 6A thermal breaker with all-pole circuit break is recommended.
- Make sure there is a differential switch with 0.03A threshold upstream of the system.
- Check that the earthing system is correctly made and connect the closure metal parts to it. Also connect the Yellow/Green wire of the automation to the earthing system.
- The automation includes an intrinsic anti-crushing device consisting of a torque control which, however, must be installed together with other safety devices.
- The safety devices (e.g.: photocells, sensitive edges, etc...) protect any dangerous areas against **Movement mechanical risks**, such as, crushing, dragging, or shearing.
- Use of at least one indicator-light is essential for each system (e.g.: FAAC LAMP MINILAMP, etc.) as well as a sign adequately fixed to the frame structure, in addition to the devices mentioned at point "16".
- FAAC accepts no responsibility regarding safety and correct operation of the automation, should components made by manufacturers other than FAAC be used in the system.
- Use only FAAC original spare parts for maintenance.
- Do not make any alterations to the components of the automation.
- The installer must supply full information regarding manual operation of the system in case of an emergency and hand to the user of the system the "User's Guide" included with the product.
- Do not allow children or other persons to stand near the product while in operation.
- Keep remote controls or any other pulse generator well away from children, to prevent the automation from being activated accidentally.
- The user must refrain from attempting to repair or adjust the system personally and should contact qualified personnel only.
- Anything not expressly provided for in these instructions is not permitted.**

1. GENERAL CHARACTERISTICS

The control unit is designed for installation in a watertight outdoor use container, which is also designed to house 2 optional batteries and a toroidal transformer, with the characteristics shown in the table below.

3. TECHNICAL SPECIFICATIONS

Supply voltage of transformer	230V~ (+ 6 % - 10 %) 50Hz
Supply voltage of control unit	22V~ (+ 6 % - 10 %) 50Hz
Power consumption	3 W
Motor max. load	2 x 70 W
Accessories max. load	24Vdc 500mA
Flashlight max. load	24Vdc 15W max
Ambient temperature	-20 ÷ +55 °C
Fuse	2
Function logics	Automatic / Stepped
Opening / closing time	Through self-learning at installation
Pause time	Through self-learning at installation
Thrust force	Two levels selected by Dip-switch
Time differences	Two levels selected by Dip-switch
Deceleration	At opening and closing during self-learning
Terminal board inputs	22 V~power supply / Battery power supply / Total opening / Opening for pedestrians / Stop / Photocells
Quick connector	For decoding - RP receivers card
Terminal board outputs	24 Vdc accessories power supply / 24 Vdc Motors Electric lock / 24 Vdc Flashlight / 24 Vdc Warning-light
Card dimensions	170 x 130 mm
Characteristics of toroidal transformer	
prim. 230V~ sec. 22V~ / 150VA / dims. Ø 105 x 40 mm	
Characteristics of optional batteries	
12V 4.5Ah / dims. 90 x 70 x 108 mm	
Characteristics of outdoor container	
305 x 225 x 125 mm - IP55	

4. INSTRUCTIONS

WARNING: To ensure people's safety, all warnings and instructions in this booklet must be carefully observed. Incorrect installation or use of the product could cause serious harm to people.

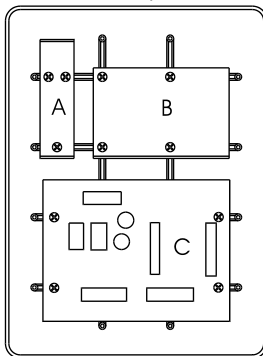
Make sure there is an adequate differential switch upstream of the system as specified by current laws, and install a thermal breaker with all-pole switch on the electrical mains. To lay electrical cables, use adequate rigid and/or flexible tubes.

Always separate connection cables of low voltage accessories from those operating at 230 V~. To prevent any interference whatever, use separate sheaths.

The maximum length of power cables between control unit and motors must not exceed 10 m, using cables of 2.5mm² diameter.

Procedure for securing components in the watertight container:

- 1) Secure the support of the toroidal transformer in position **A** by fitting three Ø3.9x6.5 self-tapping screws (supplied) in the guides of the watertight container.
N.B.: the support is sized to house a transformer (not supplied) with characteristics and dimensions specified in the table in paragraph 3.
- 2) Secure the transformer on the support with 2 clamps (supplied).
- 3) If using floating batteries, secure the relevant support in position **B** by fitting four Ø3.9x6.5 self-tapping screws (supplied) in the crossover holes of the guides of the watertight container.
N.B.: the support is sized to house two batteries (not supplied) with characteristics and dimensions specified in the table in paragraph 3.
- 4) Position the batteries on the support.
- 5) Secure the control unit in position **C** by fitting four Ø3.9x6.5 self-tapping screws (supplied) in the guides of the watertight container.



5. CONNECTIONS AND OPERATION

5.1 TERMINAL BOARD M1

22V power supply

Terminals "1-2". This is the input to which the secondary winding of the transformer, powered at 22 V 50/60 Hz, should be connected. When power is supplied by the transformer, this is signalled by the POWER LED lighting up.

Batteries

Terminals "3-4". The control unit is designed to operate with two floating batteries (optional item) with minimum characteristics as indicated on the table in paragraph 3.

When powered, the control unit keeps the batteries charged. The batteries begin to operate when the transformer no longer supplies power.

N.B.: power supplied by batteries should be considered an emergency situation – the minimum number of gate operations is about 10/15. In any event, the number of possible operations depends on the quality of the batteries, structure of gate, and time lapsing since mains power failed, etc., etc..

N.B.: observe the power polarity of the batteries

Accessories

Terminals "5-6". Output for powering external accessories (24 Vdc).

N.B.: maximum load of accessories is 500 mA.

5.2 TERMINAL BOARD M2

Gearmotor 1

Terminals "7-8". Connect the motor for leaf 1 of twin-leaf gates to power supply of 24 Vdc 70W max.

Use for gearmotor connection for single-leaf gates.

Gearmotor 2

Terminals "9-10". Connect the motor for leaf 2 of twin-leaf gates to power supply of 24 Vdc 70W max.

Do not connect for single-leaf gates.

Electric lock

Terminals "11-12". Connect an electric lock operating on 24Vdc 24W max power supply. Depending on gate structure and type of electric lock fitted, dip-switch 5 can be used to set the leaf reversing stroke enabling the electric lock to release.

N.B.: install the electric lock on the leaf on which gearmotor 1 is fitted.

Flashlight

Terminals "13-12". Use a flashlight with fixed light operating voltage of 24Vdc 15W max. We advise you to connect it before programming, as it indicates programming stages. On opening, it pre-flashes steadily for 0.5 seconds, and for 1.5 seconds at closing. If the automatic logic is on, when the gate reaches the opening gate stop, the flashlight stays on for 5 sec to inform the user it will re-close automatically. When the gate is open, the flashlight is off, and only flashes when the safety devices are in use. If the devices are in use for a long time, flashing only lasts 10 sec.

Warning-light

Terminals "14-12". Use a warning-light operating on 24Vdc 3W max. When the gate is closed, the warning-light is off, and goes on at opening, gate open and closing stages.

5.3 TERMINAL BOARD M3

OPEN-B (For pedestrian use)

Terminals "15-19". Any device (e.g. push-button, remote control, etc.) can be connected to this circuit.

By closing a contact, the circuit generates a pulse for partial opening of the gate. If the gate has two leaves, a pulse fully opens the leaf connected to gearmotor 1. If the gate has one leaf only, the pulse partly opens the leaf (50% of work time).

N.B.: an OPEN-A pulse during the pedestrian stage always has priority over that stage

N.B.: to install several pulse generators, connect the contacts in parallel.

OPEN-A

Terminals "16-19". Any device (e.g. push-button, remote control, etc.) can be connected to this circuit.

By closing a contact, the circuit generates a pulse for total opening and/or closing of the gate. Its operating mode is set by dip-switch 3 – see the relevant paragraph.

N.B.: an OPEN-A pulse during the pedestrian stage always has priority over that stage

N.B.: to install several pulse generators, connect the contacts in parallel.

Photocells

Terminals "17-19". Any safety device (e.g. photocell, sensitive strip, etc.) can be connected to this circuit. By opening a contact, the circuit protects closing motion. The status of this input is signalled by the FTO LED. It also has an effect on opening motion, depending on how dip-switch 4 was set – see relevant paragraph.

N.B.: If safety devices are not connected, fit a jumper at input. To install several safety devices, connect the NC contacts in series.

STOP

Terminals "18-19". Any device (e.g. push-button, remote control, etc.) can be connected to this circuit.

By opening a contact, the circuit stops gate movement. The status of this input is signalled by the STOP LED. The set cycle will restart only if a successive opening or closing pulse is received.

N.B.: If STOP devices are not connected, fit a jumper at input. To install several STOP devices, connect the NC contacts in series.

6. FITTING DECODING/RP RECEIVER CARDS

Installation procedure: turn off power and fit the module in container M5 inside the control unit. Then observe the radio-receiver instructions to store data on the remote-control. After the necessary data has been stored, the remote-control activates OPEN-A like any other command device.

7. SETTINGS WITH DIP-SWITCH S1

SW1	ELECTRONIC CLUTCH
	ON Maximum force, minimum sensitivity
	OFF Minimum force, maximum sensitivity
SW2	FUNCTION LOGIC
	ON Automatic
	OFF Stepped
SW3	OPERATION OF OPENING COMMAND
	ON One state only at each pulse; open, stop, close, stop, open etc.
	OFF One movement only at each pulse: open, close, open, close, etc.
SW4	OPERATION OF PHOTOCELLS
	ON Stops at opening, restarts on release, stops at closing and reverses
	OFF Stops and reverses at closing only
SW5	REVERSE PULSE
	ON Generates a pulse for 1.5 seconds at closing
	OFF Disabled
SW6	TIME DIFFERENCE BETWEEN MOTOR 1 AND MOTOR 2
	ON 2 seconds at opening, 12 seconds at closing
	OFF 2 seconds at opening, 4 seconds at closing
SW7	GATE
	ON Twin-leaf: two gearmotors connected
	OFF Single-leaf: one gearmotor connected

8. CONTROL LEDS

LED	LIGHTED	OFF
POWER - power	with transformer	on battery (if used)
FTO - photocells	photocells not covered	photocells covered
STOP - stop	command inactive	command active

N.B.: LED status shown in bold with gate closed and control unit powered.

9. PROGRAMMING

Programming of work times, deceleration and electronic clutch is executed during self-learning. At this stage, leaf movement is at slow speed.

Important: the self-learning procedure must be executed with back-up batteries (optional) disconnected.

Procedure:

- 1) Release the leaves, take them to about midway through opening travel, and then lock them.
- 2) Power up the control unit (power ON is signalled by the POWER LED).
- 3) Turn switch **S2** to **PROG**: the flashlight goes on at steady light to signal programming.
- 4) Press the push-button connected to the OPEN-A terminals or the remote-control, if already programmed. The first operation the automation performs must be CLOSING. First, the leaf connected to M2 closes, and then the one connected to M1.
- 5) If the leaves move to open, touch the two RESET pins with a screwdriver – the control unit will immediately stop the movement generated by the automation.
- 6) Cut power to the control unit, reverse polarity of the two cables powering the motors which had supplied opening drive, and repeat the operation at point 1.
- 7) After the OPEN-A command is given, the leaves move to close, until they reach the closing gate stop.
- 8) After about two seconds, the leaf connected to M1 restarts opening automatically, and after another two seconds, the leaf connected to M2 also restarts until it reaches the opening gate stops.
- 9) The control unit begins counting pause time. After the required time has elapsed, press the OPEN-A command again, and the leaf connected to M2 restarts closing. After the set time difference has elapsed, the leaf connected to M1 restarts too until it reaches the closing gate stops.
- 10) Programming is now finished. Turn switch **S2** back to **OFF** – the flashlight goes off.

10. OPERATION OF ELECTRONIC CLUTCH

This is a very important device for safety. Its setting does not alter through time, as the device is not subject to wear or setting changes.

It is active both at closing and opening. When it operates it reverses motion direction without inhibiting automatic closing if enabled.

If it operates twice in succession, it moves to STOP position, disabling any automatic command.

This is why: if the clutch operates twice, this means the obstacle is still present and any further manoeuvres could be dangerous, thus obliging the user to give an opening or closing command. In that case the control unit executes an EMERGENCY procedure as follows: full opening at slow speed up to the opening gate stop, followed by automatic closing to enable the gate stops to re-synchronise independently.

11. FUSES

FUSE	PROTECTING
F1 = F 3.15A 250V 5x20	Logic / Accessories output
F2 = T 10A 250V 5x20	Motor

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- The present manual is produced in 100% recycled paper
- Respect the environment. Dispose of each type of product packaging material (card, polystyrene) in accordance with the provisions for waste disposal as specified in the country of installation.

FAAC der Umwelt zuliebe

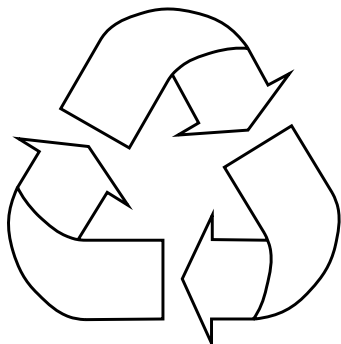
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